

Takeoff Rulebook (R1–R15)

Load this in Step 5. Apply these rules in order to each room in the inventory. Later rules can ADD signs but do not override earlier assignments unless explicitly noted.

For every sign assigned, record which rule triggered it. This is the audit trail.

Room ID rules (R1–R7)

R1 — Room ID default

Condition: Room is occupied AND NOT (corridor/hall/lobby) AND NOT vehicle bay AND NOT unoccupied MEP. **Assign:** 1 × Room ID (Plaque A) to the **Room ID** column.

Example from Example-1: Room 109 POLICE CHIEF, occupant_load = 1, is_mep_unoccupied = false, is_corridor = false → 1 × Room ID.

R2 — Room ID with insert

Condition: **is_variable_use** = true. **Assign:** Use "Room ID w/insert" instead of plain Room ID. Quantity = 2 if room has ≥2 egress doors OR dual function (e.g., training + EOC), else 1.

Example from Example-1:

- Room 135 DE-ESCALATION TRAINING: is_variable_use=true, dual-function (training + observation) → **2** Room ID w/insert
- Room 138 TRAINING/COMMUNITY/EOC: is_variable_use=true, triple-function → **2** Room ID w/insert

R3 — Multi-entry large rooms

Condition: Room has ≥3 man-doors (pedestrian doors, not vehicle doors). **Assign:** Room ID quantity = number of man-door entries.

Example from Example-1: Room 146 APPARATUS has 3 man-doors for firefighter access → **3** × Room ID.

Note: This rule requires inspecting the floor plan to count doors. If you can't determine the count, default to 1 and flag in "Questions for verification".

R4 — Corridor/hall exclusion

Condition: `is_corridor_or_hall` = true. **Assign:** NO Room ID. Corridors get Office Directory instead per R14.

Example: Room 140 HALL gets 0 Room ID, but gets 1 Office Directory (see R14).

R5 — Vehicle bay exclusion

Condition: `is_vehicle_bay` = true. **Assign:** NO Room ID, unless explicitly in scope (confirm with client).

Exception noted in Example-1: APPARATUS got 3 Room IDs via R3. This is project- specific; confirm whether apparatus bays are included for new projects.

R6 — Unoccupied MEP exclusion

Condition: `is_mep_unoccupied` = true AND room is on mezzanine. **Assign:** NO Room ID.

Example from Example-1: Mezzanine rooms (Server 302, Town Data Center 303, Mechanical Attic 304) are excluded per R15 mezzanine rule.

For non-mezzanine MEP rooms (like Fire Sprinkler 141, Emerg Director 142 on L1): these GET a Room ID because they are on an occupied floor and the rooms themselves have people who enter them regularly.

R7 — Dorm / sleeping rooms

Condition: Room is a dormitory, bunk room, or assigned sleeping room. **Assign:** Standard Room ID (Plaque A), NOT Room ID w/insert. Dorms are assigned to specific occupants, not variable-use.

Example from Example-1: Rooms 153 DORM, 154 DORM → 1 Room ID each (not w/insert).

Restroom rule (R8)

R8 — Restroom plaque selection

Condition: `is_restroom` = true. **Assign:** 1 × Restroom plaque. Plaque variant selected by:

Attributes	Plaque variant
Public-facing AND gendered women	B (Women H/C — accessible)
Public-facing AND gendered men	C (Men H/C — accessible)
Staff-only AND gendered women	F (Women Staff)
Staff-only AND gendered men	E (Men Staff)
Unisex / single-user (no gender in name)	Accessible variant (default B or project's unisex plaque)
Name is just "TOILET" with no gender	Accessible variant (single-user default)

Examples from Example-1:

- Room 102 PUBLIC TOILET (public, unisex) → 1 Restroom (accessible single-user)
 - Room 108 TOILET (inside staff zone, unisex) → 1 Restroom (accessible)
 - Room 119 DET SHOWER (staff only, unisex) → 1 Restroom
 - Room 133 MALE STAFF (staff, male) → 1 Restroom (Plaque E)
 - Room 134 FEMALE STAFF (staff, female) → 1 Restroom (Plaque F)
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EXIT rule (R9)

R9 — EXIT plaque placement

Condition: Varies. Assign **1 × EXIT plaque** at:

1. Each exit-discharge vestibule (**is_vestibule** AND leads to exterior)
2. Public lobby with an exit door (**is_public_facing** AND has egress door)
3. Each required exit from an assembly room — quantity per IBC Table 1006.2.1:
 - Occupant load < 50 (Business): 1 exit
 - Occupant load ≥ 50 OR assembly use: 2 exits minimum

Examples from Example-1 Level 1:

- Room 100 VEST (entry vestibule) → 1 EXIT
- Room 101 LOBBY (public, has exit door) → 1 EXIT
- Room 138 TRAINING/COMMUNITY/EOC (assembly, occ_load > 50) → **2** EXITs
- Room 157 VEST (rear vestibule with double exit) → **2** EXITs

- **L1 total: 4 EXITS ✓** (matches human takeoff subtotal)
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CAPACITY rule (R10)

R10 — Max Occupancy sign

Condition: Room is assembly use. **Assign:** 1 × Capacity insert per separable assembly space.

Triggers (any one):

- **occupancy_group** is A-2 (Assembly Unconcentrated — meeting, conference, dining)
- **occupancy_group** is A-3 (Assembly Concentrated — theater, auditorium)
- **occupant_load** ≥ 50 AND use is assembly-like

Examples from Example-1:

- Room 135 DE-ESCALATION TRAINING → per takeoff, NO capacity sign (low occupant load per A-003). Verify per project.
 - Room 138 TRAINING/COMMUNITY/EOC → **2** Capacity signs (assembly + EOC are separate)
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Stair rules (R11)

R11 — Stair plaque set per stair per level

For each stair, at each level it serves:

- **Stair (Corridor):** 1 per corridor entry door into the stair on that level
- **Stair (Landing):** 1 inside the stair at that level's landing

Count corridor entries by inspecting the floor plan for door swings into the stair.

Example from Example-1 Level 1:

- STAIR 1: 1 corridor + 1 landing = 1, 1
- STAIR 2: 2 corridor (has two entries on L1) + 1 landing = 2, 1
- **L1 stair totals: 3 corridor + 2 landing ✓**

Example from Example-1 Level 2:

- STAIR 1: 1 corridor + 1 landing
 - STAIR 2: 1 corridor + 1 landing
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Elevator rules (R12)

R12 — Elevator signs

Per elevator (not per floor):

- **1 × "In Case of Fire..."** plaque (mounted at the main floor near call buttons)

Example from Example-1:

- 1 elevator × 1 "In Case of Fire" = 1 total ✓
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Evacuation map rule (R13)

R13 — Evacuation Map placement

Assign 1 × Evacuation Map at:

- Each elevator lobby (at every floor)
- Each main building lobby / public entry
- Key circulation decision points (per life safety sheet if shown)

Examples from Example-1 Level 1:

- Elevator lobby: 1 Evac Map
 - Main LOBBY 101: 1 Evac Map
 - **L1 total: 2 Evac Maps ✓**
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Office directory rule (R14)

R14 — Office Directory placement

Assign 1 × Office Directory at:

- Main public-facing hall/lobby per wing/department zone per level
- Typically 1 per major building wing on each level

Examples from Example-1:

- L1: LOBBY 101 (1) + HALL 140 (1) = **2**
 - L2: HALL 201A (1) + CORR 221 (1) = **2**
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Mezzanine rule (R15)

R15 — Mezzanine exclusion

Condition: Level is a mezzanine AND contains only unoccupied MEP/IT spaces + stair/elevator access. **Assign:** NO signs on mezzanine.

Exception: If stair/elevator opens onto an OCCUPIED mezzanine space, include stair/elevator signs at that level per R11 and R13.

Example from Example-1: Mezzanine has Server Room 302, Town Data Center 303, Mechanical Attic 304 — all unoccupied MEP/IT. Mezzanine is excluded from takeoff.

Rule application order

Process each room through the rules in numerical order. Record results as you go:

Room 138 TRAINING/COMMUNITY/EOC (L1):

- R2 triggered (is_variable_use) → 2 × Room ID w/insert
- R9 triggered (assembly, occ_load > 50) → 2 × EXIT

- R10 triggered (A-2 occupancy) → 2 × Capacity

Total: 2 in "Room ID w/insert" + 2 in "Exit" + 2 in "Max. Occupancy"

Building Type Rules (R16-R17) and Trait-Based Modifications

R16 — Unit / Room Number plaque (for buildings with assigned units)

Applies to: Residential, Senior Living, Student Housing, Hotels **Condition:** `room_name` starts with "UNIT", "ROOM" (hotel), "SUITE", or "APT" **Assign:** 1 × Unit/Room # plaque in a **separate column** from Room/Service Rm ID. These are assigned dwelling or guest spaces, NOT variable-use rooms.

R17 — Multi-building projects

Condition: Project contains multiple buildings (separate addresses, drawing series, or explicit building labels on the plan index). **Action:** Each building gets its own independent takeoff with its own level structure, room inventory, and subtotals.

Building Type Trait System

Rather than hardcoding rules per building type, the skill uses **6 trait dimensions**. Each building type maps to a specific combination of trait settings. The rules engine reads these traits to determine which rules apply and how.

Trait definitions

Trait 1: `has_unit_numbers` — Do rooms get a separate Unit/Room # plaque?

- `true`: Residential, Senior Living, Student Housing, Hotels, Dormitories
- `false`: All others

Trait 2: `mep_room_id_policy` — Do MEP rooms (Mechanical, Electrical, etc.) get Room ID?

- `all`: Every MEP room with a door gets Room ID. (Residential, Hotels, Retail, Warehouse, Factories)
- `occupied_only`: Only MEP rooms with occupant load > 0. (Commercial Office, Labs, Banks, Government)

- **minimal**: MEP rooms rarely get Room ID; typically excluded unless scoped. (Schools, Hospitals)

Trait 3: has_assembly_rules — Do Max Occupancy (R10) and assembly EXIT (R9) rules apply?

- **true**: Commercial Office, Schools, Hospitals (waiting areas), Government, Churches, Restaurants, Retail (spaces with occupant load ≥ 50), Hotels (conference/ballrooms)
- **false**: Residential, Warehouse, Factories (no assembly occupancy)

Trait 4: has_directory — Does R14 (Office Directory) apply?

- **true**: Commercial Office, Banks, Government, Medical Office, Labs
- **false**: Residential, Schools, Hospitals, Hotels, Retail, Warehouse, Factories, Churches

Trait 5: elevator_sign_mode — How are elevator signs counted?

- **per_building**: 1 × "In Case of Fire" per elevator total. (Commercial, Schools, Banks, Labs, Retail, Warehouse)
- **per_level**: 1 × Elevator ID per elevator per level it serves. (Residential, Hotels, Hospitals)

Trait 6: stair_column_mode — How are stair signs reported?

- **separate**: Stair (Corridor) and Stair (Landing) as separate columns. (Commercial, Schools, Hospitals)
- **combined**: Stairs (Corridor + Landing) as one column, value = 2 per stair per level. (Residential, Hotels)

Building type → trait mapping

Building Type has_units mep_policy assembly directory elevator_mode stair_mode

Residential	true	all	false	false	per_level	combined
Commercial Office	false	occupied_only	true	true	per_building	separate
School / University	false	minimal	true	false	per_building	separate
Hospital / Healthcare	false	minimal	true	false	per_level	separate

Hotel / Hospitality	true	all	true*	false	per_level	combined
Retail / Restaurant	false	all	true	false	per_building	separate
Warehouse / Factory	false	all	false	false	per_building	separate
Lab / Research	false	occupied_only	false	true	per_building	separate
Bank / Financial	false	occupied_only	false	true	per_building	separate
Government / Civic	false	occupied_only	true	true	per_building	separate
Church / Worship	false	occupied_only	true	false	per_building	separate
Senior Living	true	all	true*	false	per_level	combined
Mixed-Use	per-floor	per-floor	per-floor	per-floor	per-floor	per-floor

*Hotels/Senior Living: assembly rules apply only to conference rooms, ballrooms, dining halls — NOT to individual guest/resident rooms.

Building type detection (Phase 1)

Detect building type from room names and project metadata, in priority order:

1. **If project name/metadata explicitly states type** → use it
2. **If >30% of rooms named "UNIT xxx"** → Residential
3. **If rooms named "ROOM xxx" + "LOBBY" + "FRONT DESK"** → Hotel
4. **If rooms include "PATIENT", "NURSE", "EXAM", "OR", "ICU"** → Hospital
5. **If rooms include "CLASSROOM", "GYMNASIUM", "CAFETERIA", "PRINCIPAL"** → School
6. **If rooms include "NAVE", "SANCTUARY", "ALTAR", "PARISH"** → Church
7. **If rooms include "CLEAN ROOM", "FUME HOOD", "BSL-"** → Lab
8. **If rooms include "VAULT", "TELLER", "SAFE DEPOSIT"** → Bank
9. **If rooms include "ASSEMBLY LINE", "LOADING DOCK", "SHIPPING"** → Warehouse/Factory
10. **If rooms include "MERCHANDISE", "FITTING", "POS", "STOCK ROOM"** → Retail
11. **If IBC occupancy group is specified:** R-1/R-2 → Residential, B → Commercial, E → School, I-2 → Hospital, A → Assembly, S → Warehouse, H → Hazardous, F → Factory
12. **Default:** Commercial Office (most common project type)

How traits modify rules R1-R15

R1 (Room ID):

- `mep_policy=all`: ALL rooms with doors get Room ID (no MEP exclusion)
- `mep_policy=occupied_only`: MEP rooms only if `occupant_load > 0` (R6 applies)
- `mep_policy=minimal`: MEP rooms excluded unless explicitly in scope

R2 (Variable-use inserts):

- Only relevant when `has_assembly_rules=true` (variable-use rooms exist)
- Hotels: conference rooms get inserts; guest rooms do NOT

R6 (MEP exclusion):

- Applies when `mep_policy=occupied_only` or `mep_policy=minimal`
- Does NOT apply when `mep_policy=all`

R9 (EXIT):

- Always applies to vestibules at exit paths
- Assembly-based EXIT (2 per assembly room with ≥ 50 occ) only when `has_assembly_rules=true`

R10 (Max Occupancy):

- Only applies when `has_assembly_rules=true`
- For Hotels/Senior Living: only on conference/assembly spaces, NOT guest rooms

R11 (Stairs):

- `stair_mode=separate`: Report as two columns (Corridor, Landing)
- `stair_mode=combined`: Report as one column, value = corridor + landing per stair

R12 (Elevator):

- `elevator_mode=per_building`: 1 × "In Case of Fire" per elevator total
- `elevator_mode=per_level`: 1 × Elevator sign per elevator per level it serves

R13 (Evacuation Map):

- `elevator_mode=per_building`: Evac at elevator lobbies + main entries

- `elevator_mode=per_level`: Evac on occupied floors only (not garages/parking)

R14 (Office Directory):

- Only applies when `has_directory=true`

R16 (Unit/Room #):

- Only applies when `has_unit_numbers=true`

Special sign types by building type

Some building types have additional sign types beyond the base R1-R15:

Hospitals/Healthcare:

- Nurse Station ID sign
- Isolation Room warning sign
- Radiation warning sign (if imaging department)
- No Smoking / Oxygen in Use

Labs/Research:

- Lab Safety sign (chemical hazard, BSL level)
- Fume Hood warning
- Emergency Shower / Eyewash location

Warehouse/Factory:

- Hazmat / Chemical Storage warning
- Loading Dock identification
- Forklift Traffic warning
- Emergency Assembly Point

Schools:

- Classroom Number (sometimes separate from Room ID)
- Capacity sign for gymnasium, auditorium, cafeteria

Hotels:

- Room Number plaque (same as Unit # but labeled "Room")
- Do Not Disturb / Privacy sign provisions (usually not in ADA takeoff)

- Pool Rules / Fitness Rules (if amenity spaces)

Banks:

- Vault identification
- Safe Deposit identification
- Security notice signs

Churches:

- Assembly capacity for sanctuary, fellowship hall
- Fire exit for large worship spaces

These special types should be flagged in "Questions for verification" if detected but not explicitly in the plaque schedule. The base R1-R15 + R16-R17 cover ~90% of signs for any building type; special types are project-specific scope additions.

Column schema — dynamic, not hardcoded

Instead of selecting a fixed column schema per building type, the output columns should be **dynamically determined** by which sign types were actually assigned:

1. After running all rules, collect the set of unique **sign_type** values across all rooms
 2. These become the output columns, in this standard order: Room/Service ID → Unit/Room # → Room ID w/insert → Restroom → Exit → Max Occupancy → Stair (Corridor) → Stair (Landing) OR Stairs (combined) → Elevator → Evacuation Map → In Case of Fire → Office Directory → [any special types]
 3. Omit columns where every row is blank (e.g., no Max Occupancy in residential)
 4. This means the output automatically adapts to the building type without hardcoding
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Edge cases

What if a room matches multiple rules for the same column?

The rules are additive for quantity but selective for variant. For example:

- A large training room with 3 egress doors: R2 gives 2 Room IDs w/insert + R9 gives 3 EXITS

- R3 (multi-entry) can increase Room ID count beyond the R2 default

What if the plaque schedule doesn't have a specific plaque type?

Default to the closest functional match and flag in "Questions for verification". Example: if no "Staff Restroom" plaque is shown, default to the H/C (accessible) variant.

What if a room has no clear category?

Assign nothing, flag in "Questions for verification", and proceed. Do not guess.

What if the project has both residential and commercial spaces (mixed-use)?

Apply residential rules to residential floors/units, commercial rules to commercial floors/spaces. Detect per-floor: if a floor has UNIT rooms, treat it as residential; if it has office/retail rooms, treat it as commercial. The output columns should be the superset of both. Flag the mixed-use classification in "Decisions & Assumptions."

What if the building type is unclear?

Default to Commercial Office (the most common and most conservative rule set). Flag the assumption and let the user correct via the training feedback loop.